To: Dr. Naqaa Abbas, ENGL 220
From: Team Cocktail—Aazim Rafeeq, Hiba Syed, Kel Doliente, Khaldoun Saadeddine, and Youssef Abou Ali
Date: Saturday, April 10, 2026
Subject: Technical Report Draft - Lack of Pedestrian Walkability in Education City, Qatar

TECHNICAL REPORT DRAFT

I. Abstract

Education City (EC) in Qatar spans 12 square kilometers and hosts several world-class universities, research institutions, and cultural landmarks. Despite EC's design as a walkable academic hub, most of its population relies on private vehicles for short-distance travel. Our team evaluated four potential solutions: The Park and Plant tree-planting initiative, cool pavement coatings, climate-controlled enclosed walkways, and solar-powered misting systems. A weighted decision matrix was used across four criteria: thermal comfort, sustainability, coverage and scalability, and feasibility. Park and Plant achieved the highest score, offering natural cooling through native tree shade, carbon sequestration, and campus-wide scalability without new infrastructure. The recommendation aligns with QNV 2030's pillars of environmental sustainability and human development, leveraging native tree species such as Sidra and Ghaf to provide natural shade, improve air quality, sequester carbon, and create a more walkable campus environment.

II. Introduction

Education City is a major project of the Qatar Foundation, designed to support academic excellence and cross-institutional collaboration within a 12-square-kilometer campus in Doha. It includes campuses of internationally recognized universities, such as Hamad Bin Khalifa University, Texas A&M University at Qatar, Carnegie Mellon University in Qatar, and several more, alongside research centers, Oxygen Park, the Qatar National Library, and the Qatar Foundation headquarters. Its urban development plan was designed with pedestrian accessibility in mind, incorporating a tram system with 24 stations to ease movement across the campus (Al-Mohannadi et al., 2023).

However, Education City's walkability has not met design expectations. Qatar's extreme summer climate, with daytime temperatures averaging 45°C and humidity levels ranging from 40% to 60%, results in conditions that restrict outdoor pedestrian activity (Visit Qatar, 2026). Research by Al-Mohannadi et al. (2023) found that only 17% of EC's students, staff, and visitors walk between establishments, while the vast majority depend on private vehicles or shuttle services. The effects of this car dependency are significant and include increased carbon emissions, reduced physical activity, and weakened community interaction across campuses.

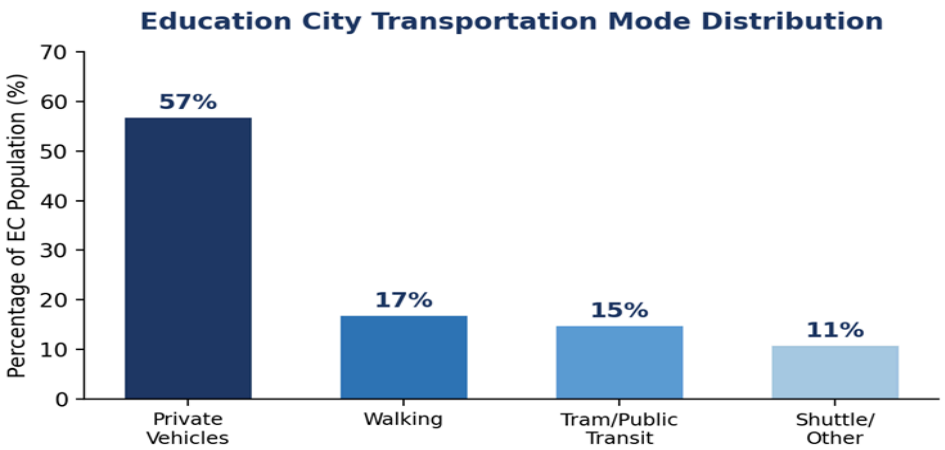

Figure 1: Transportation mode distribution among the Education City population (Al-Mohannadi et al., 2023).

This challenge directly contradicts Qatar's national policy objectives. The Third National Development Strategy (NDS3) identifies transportation as a highly important sector for emissions reduction and calls for creating more recreational spaces, improving transport accessibility, and promoting sustainable transportation (National Planning Council, 2024). Furthermore, Qatar's National Master Plan recommends shading 60–70% of pedestrian walkways to facilitate walking in hot-arid conditions (Ministry of Municipality and Environment, n.d.). Walkability is recognized as a crucial element of proper sustainable urban design, reducing car dependency, lowering emissions, improving public health, and strengthening community life (Baobeid et al., 2021).

The objective of this report is to investigate the primary causes of poor pedestrian walkability in Education City, evaluate potential solutions, and recommend the most effective and practical approach for the campus setting. The report relies on academic literature, institutional data, and established engineering evaluation guidelines to support its assessment and suggestions.

III. Methodology

This study was performed using a multi-method approach for research by combining different sources and institutional analysis, and has also used different types of engineering evaluation tools to better understand the issue of walkability in Education City and to identify the solutions carefully.

1. Literature Review

To start with, the team reviewed a number of academic sources that were related to urban design, walkability, and thermal comfort in the weather of Qatar. Databases such as Google Scholar, Scopus, and the HBKU library database were used to find peer-reviewed articles. Studies performed by Al-Mohannadi et al. (2023) were useful in the understanding of transit-oriented development, proper urban planning, and improved mobility in the EC, while the studies by Middel et al. (2023) and Wang et al. (2024) gave quantitative data on different cooling strategies. Additionally, national reports like the National Planning Council's (2025) report on Qatar's economic transition talk about how the country is working towards sustainability and improving its infrastructural development, which is also connected to improving pedestrian environments. A total of 15-20 peer-reviewed articles and institutional reports were used. Some of the key search terms used were "urban walkability in hot climates," "thermal comfort pedestrian Qatar," and "shaded walkways effectiveness." " The studies conducted in hot and arid regions were given priority, particularly within the Gulf and Qatar. Using both academic and policy sources helped make sure that the research was strong both in terms of theory and real-world planning.

2. Institutional and Contextual Analysis

The study also focused on the initiative that has already been implemented in the education city to understand what is already being done to improve walkability. Particularly, the park and plant initiative was researched as an example of how local plants can help reduce heat and improve walkability. The initiative for the Quranic botanic garden also supported this by showing how sustainable planting strategies are being implemented in Qatar. Additionally, some informal observational data were collected by the team over several weeks regarding the daily movement across EC. These observations were focused on walking distances, thermal discomfort, and shade availability. These observations, along with literature findings, helped ensure that the analysis reflects real user experience.

3. Engineering Evaluation Tools

For the comparison of all the possible solutions in a structured manner, a weighted decision matrix was used to rank the solutions for more accuracy. Criteria such as thermal comfort (30%), sustainability (30%), coverage and scalability (25%), and feasibility (15%) were ranked. Each solution was then scored on a 1-5 scale, with the existing EC canopies being used as the reference. The final ranking was then calculated by multiplying each score by its weight and adding all the results up. Using this method helped the team make an accurate decision instead of just depending on assumptions and personal opinions.

IV. Results

Our research confirmed that Education City's walkability problem is driven by a combination of extreme heat, insufficient shade, and a campus layout that favors car use over walking. With temperatures regularly exceeding 40°C from June through September and humidity levels between 40% and 60%, walking between buildings becomes physically exhausting. Only 17% of EC's population walks between facilities, while 57% prefer using their own cars (Al-Mohannadi et al., 2023). Although the campus tram provides 24 stations, many walkway segments between tram stops and building entrances remain unshaded, which limits how useful the tram actually is as a pedestrian mobility tool (Luomi et al., 2014). Gulf countries have historically been designed around cars, with wide roads and minimal space designed for walking, and Education City reflects this pattern despite its original vision as a walkable campus.

After reviewing these circumstances, our team developed four potential solutions. These fall into two broad approaches: nature-based cooling and engineered cooling systems.

Solution 1: Park and Plant Initiative

Park and Plant is Qatar Foundation's existing tree-planting program, which aims to line EC walkways with native species such as Sidra and Ghaf trees to create continuous shaded corridors. These trees block direct solar radiation while still allowing airflow, which research has shown to be more effective than full coverage. Al Tahir et al. (2025) found that 75% canopy shading provides better airflow and cooling than complete coverage. The initiative started with 2,500 trees and has a target of 20,000, supported by community engagement programs like "Adopt a Tree" and corporate sponsorship through "Green Lungs" (Qatar Foundation, n.d.). A comparable study conducted on a university campus in Dubai showed that planting Ghaf and Acacia trees reduced the annual number of hours classified as "warm/hot" from 1,291 to fewer than 300 (Naboni et al., 2018). However, native trees require several years to reach full canopy maturity, and irrigation and seasonal pruning represent ongoing maintenance needs.

Figure 2: Ghaf Tree in arid climate (Plant 2018)

Solution 2: Cool Pavement Coatings

Cool pavement coatings are reflective layers applied directly over existing walkway surfaces to increase their surface reflectivity, reducing the amount of solar heat the pavement absorbs and radiates back to pedestrians. Studies conducted in Phoenix showed that these coatings can lower surface temperatures by 2.6–4.9°C compared to conventional materials used for surfaces (Middel et al., 2023). The coatings

Figure 3: Cool pavement coating application (WHYY 2019)

are relatively affordable and can be applied with minimal construction, making them one of the easiest solutions to implement. However, there is a significant trade-off: reflective pavements may increase mean radiant temperature at the pedestrian level by 2–3°C, meaning walkers could feel hotter despite the ground being cooler (Middel et al., 2023). The coatings also degrade over time due to UV light exposure, dust, and foot traffic, requiring periodic reapplication (Bloomberg Associates, 2019). Research further suggests that cool pavements are most effective when paired with overhead shade, but combining both may not be cost-effective (Ghoniem et al., 2025)

Solution 3: Climate-Controlled Enclosed Walkways

Enclosed and air-conditioned walkways provide complete thermal protection regardless of the surrounding conditions, offering the strongest thermal comfort of any solution and making them especially helpful for vulnerable groups such as children, the elderly, and migrant workers who may lack access to private vehicles. Qatar already has working previous examples of this approach, including an air-conditioned pedestrian bridge in the Industrial Area (Qatar Living, 2015) and the elevated enclosed bridge connecting the Doha metro station to the Mall of Qatar. However, the energy demands of cooling large enclosed structures in a climate where outside temperatures exceed 45°C are substantial, likely contradicting the sustainability goals unless powered totally by renewable energy. In addition, effective enclosed networks must connect heavily populated locations to justify their high cost, a condition that Education City's spread-out layout does not meet. Construction costs are high, and enclosed environments may also reduce the social spirit associated with open public spaces (Litman, 2023).

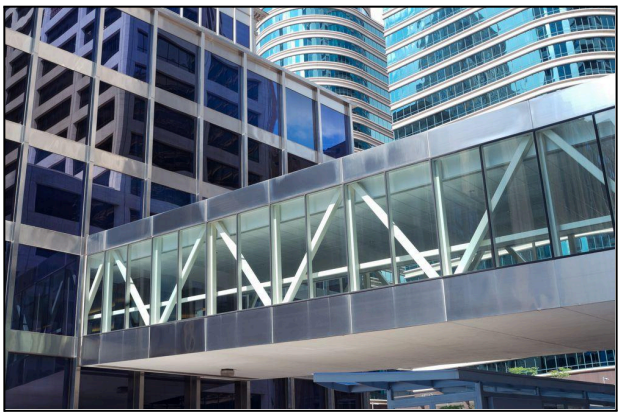

Figure 4: External view of an enclosed glass-and-steel skyway bridge connecting urban commercial buildings (Treehugger/Joe Ferrer).

Solution 4: Solar-Powered Misting Systems

This solution suggests replacing existing EC canopy structures with high-pressure misting nozzles powered by rooftop solar power panels, which produce tiny droplets that evaporate quickly, absorbing heat from the surrounding air and lowering pedestrian-level temperatures. Smart thermostatic sensors would activate the system only when the temperature exceeds a set threshold, conserving both water and energy. Research by Wang et al. (2024) shows that shade combined with misting reduces the global thermal climate index by an average of 12.3°C, a much greater impact than shade alone, which reduces it by 3.4°C. Solar-powered aerial misting has been used in similar GCC outdoor environments in Saudi Arabia and at Expo 2020 Dubai, proving its regional practicality (ResearchGate, 2013). However, misting effectiveness decreases greatly at humidity levels above 60–70% (Wang et al. 2024), and Qatar's summer humidity levels often reach 40–60%, approaching the acceptable limit. Coverage is also limited to existing canopy areas, leaving exposed walkway sections untreated. The nozzles require regular maintenance to prevent clogs and jams in Qatar's rough water conditions.

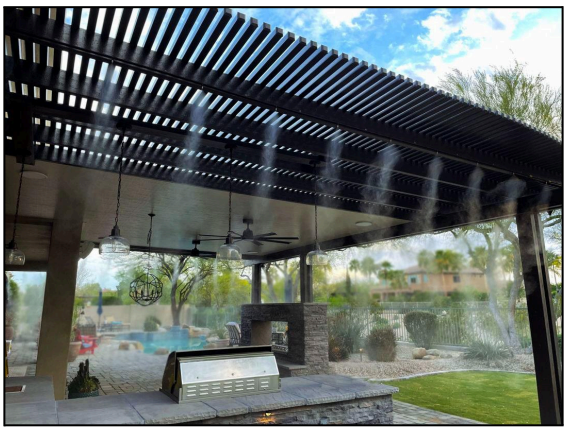

Figure 5: Misting system on a patio. Adapted from "SolarTech", https://solartechonline.com/misting-systems/

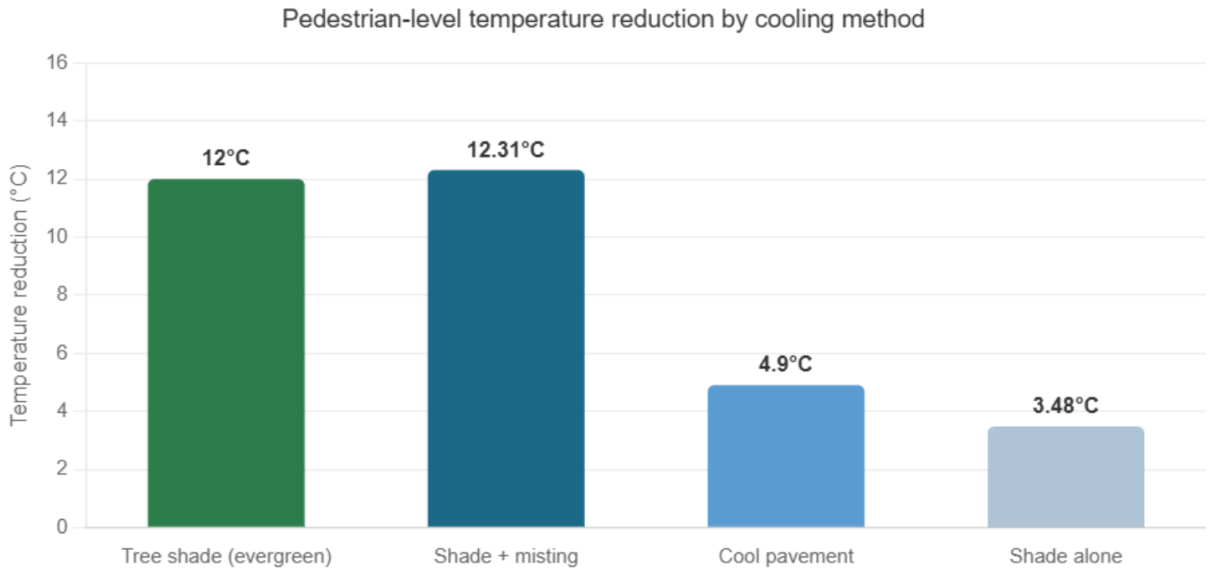

Figure 6: Pedestrian-level temperature reduction by cooling method, compiled from multiple sources (Communications Earth & Environment, 2024; Middel et al., 2023; Wang et al., 2024)

V. Analysis

The weights assigned to each criterion were determined based on their importance in QNV 2030 and logical reasoning. For instance, thermal comfort was given greater weight due to its relevance to the problem being assessed, while sustainability was assigned equal importance due to its significance in NDS3 (National Planning Council, 2024). The prevailing canopies around EC were selected as the reference concept, and solutions were evaluated on a 1–5 scale relative to this benchmark, supported by the cited literature. A score of 1 indicates a solution significantly worse than the reference, 3 indicates equal performance, and 5 indicates a significantly better outcome (American Society for Quality, 2026). Each rating was then multiplied by its corresponding percentage weight, and the resulting weighted scores were summed. The concept with the highest total score was ranked first. Based on this analysis, the Park and Plant solution was identified as the most effective, as it achieved the highest overall score.

Concepts											
Selection Criteria	**Weight**	**Park and Plant**		**Cooling and Reflective Coatings**		**Closed Air-conditioned Walkways**		**Solar-Powered Misting Systems**		**EC Canopies (Reference)**	
Customer Needs	**Σ = 100%**	**Rating**	**Weighted Score**	**Rating**	**Weighted Score**	**Rating**	**Weighted Score**	**Rating**	**Weighted Score**	**Rating**	**Weighted Score**
Thermal Comfort	30%	3	0.9	1	0.30	5	1.50	4	1.20	3	0.90
Sustainability	30%	5	1.5	2	0.60	1	0.30	4	1.20	3	0.90
Coverage & Scalability	25%	4	1.00	5	1.25	2	0.50	3	0.75	3	0.75
Feasibility	15%	4	0.60	4	0.60	1	0.15	3	0.45	3	0.45
Total Score		4.00		2.75		2.45		3.60		3.00	

Table 1. Weighted Decision Matrix for Concept Scoring

The comparison of the thermal comfort performance among the solutions highlights their varying capacity to reduce pedestrian-level heat exposure. Park and Plant scores a 3 (equal to reference) because while mature tree canopy can reduce temperatures by up to 12°C (Communications Earth & Environment, 2024), newly planted trees require 3–5 years to establish meaningful shade coverage, meaning immediate cooling matches the current canopy infrastructure. In comparison, enclosed walkways score the highest at 5 because they provide complete climate control regardless of external conditions (Litman, 2023). Solar-powered misting systems score a 4, as shade combined with misting reduces the Universal Thermal Climate Index by an average of 12.31°C, though effectiveness declines at humidity above 60–70% (Wang et al., 2024). Cool pavement coatings score the lowest at 1, because while they lower surface temperatures by 2.6–4.9°C, they may actually increase mean radiant temperature at the pedestrian level by 2–3°C, potentially worsening thermal discomfort for walkers (Middel et al., 2023).

As for sustainability, Park and Plant score the highest at 5 because trees actively sequester carbon dioxide, filter particulate matter, reduce the urban heat island effect, and improve biodiversity—directly supporting NDS3's target of 25% greenhouse gas reduction by 2030 (National Planning Council, 2024). Unlike engineered systems that consume energy, trees generate compounding environmental benefits as they mature. Solar misting scores a 4 because its solar-powered operation produces zero operational emissions, though it still consumes water—a scarce resource in Qatar (ResearchGate, 2013). Cool pavement coatings score a 2 because, while they reduce surface heat, they offer no ecological benefit and require periodic chemical reapplication that generates waste (Bloomberg Associates, 2019). Enclosed walkways score the lowest at 1, as maintaining below-ambient temperatures in large enclosed structures using air-conditioning systems requires significant energy consumption, directly conflicting with sustainability goals unless powered entirely by renewable sources (Litman, 2023).

For coverage and scalability, cool pavement coatings score the highest at 5 because they can be applied rapidly to any existing pavement surface across the entire campus with minimal structural modification (Bloomberg Associates, 2019). Park and Plant scores a 4 because tree planting can be phased across the full pedestrian network — the initiative already has a target of 20,000 trees and community programs like "Adopt a Tree" and "Green Lungs" that enable continuous expansion (Qatar Foundation, n.d.). Solar misting scores a 3 (equal to reference) because coverage is limited to existing canopy footprints, leaving uncovered walkway segments unaddressed without further canopy extensions. Enclosed walkways score the lowest at 2 because effective networks require connecting densely populated locations (Litman, 2023), and Education City's dispersed layout makes campus-wide enclosure impractical and prohibitively expensive.

Finally, for feasibility, Park and Plant and cool pavement coatings both score a 4 because they can be implemented within Education City's existing infrastructure without major construction. Park and Plant holds a decisive advantage here: it is an already operational Qatar Foundation initiative with established staffing, budgets, governance, and community engagement programs, no new procurement processes or stakeholder agreements are required (Qatar Foundation, n.d.). CMU-Q has already planted 109 trees, and Qatar Chamber has formally endorsed the initiative. Solar misting scores a 3 because while it retrofits existing canopy structures, it introduces new maintenance complexities, including nozzle blockages from Qatar's hard water and bacterial buildup concerns. Enclosed walkways score the lowest at 1, as they require heavy construction, high capital investment, and ongoing energy costs that are difficult to justify for an open campus layout (Litman, 2023).

After tallying up the scores on the decision matrix in Table 2, Park and Plant scored the highest at 4.00 points. It excels in sustainability and performs strongly across all other criteria, making it the most balanced and context-appropriate solution. Solar misting scored second at 3.60, performing well in thermal comfort and sustainability but limited by coverage and maintenance concerns. The EC canopy reference scored 3.00, as meant to be. Cool pavement coatings scored 2.75, excelling in coverage but critically weak in thermal comfort. Enclosed walkways, with a score of 2.45, achieve the best thermal comfort but are severely hindered by poor sustainability, limited scalability, and high costs.

VI. Discussion of Recommendation

After evaluating all four solutions using the weighted decision matrix, the research confirms that expanding Qatar Foundation's Park and Plant initiative is the most effective, sustainable, and feasible response to Education City's walkability challenge. Park and Plant achieved the highest weighted score (4.00/5.0), outperforming solar misting (3.60), the current canopy reference (3.00), cool pavement coatings (2.75), and enclosed walkways (2.45).

Proven and Already Operational

Park and Plant is an existing Qatar Foundation initiative with established staffing, budgets, and community programs. Over 2,500 trees have already been planted with a target of 20,000, supported by "Adopt a Tree" and "Green Lungs" sponsorship programs (Qatar Foundation, 2026). CMU-Q planted 109 trees during Sustainability Week, and the Qatar Chamber has formally endorsed the initiative. This existing framework eliminates the need for new governance structures or procurement processes, giving it a feasibility advantage that no other solution offers.

Evidence of Effectiveness

Urban trees can reduce pedestrian temperatures by up to 12°C, effective in arid climates (Communications Earth & Environment, 2024). On a Dubai campus, Ghaf and Acacia planting cut “warm/hot” hours by over 75% (Naboni et al., 2018), while 75% canopy coverage optimizes airflow and cooling (Al Tahir et al., 2025). These show that strategic tree planting significantly improves outdoor comfort and supports walking in hot-arid environments (Baobeid et al., 2021).

Promotes Walkability

Shaded pathways in Education City promote walking, reducing reliance on carbon-intensive transport while enhancing green space interaction. This supports Qatar’s sustainability goals (Patriche et al., 2023) and serves as an effective cooling strategy that helps lower vehicle emissions, which reached 34,239.15 tCO_2e in 2024 (Qatar University, 2024). Overall, improving walkability supports sustainable urban development through both environmental and social benefits (Ali et al., 2023).

Native Species and Implementation Plan

Sidra and Ghaf are drought-tolerant, low-water species suited to Qatar’s climate. The Qur'anic Botanic Garden at Education City maintains over 6,800 native plants for planting (QBG, n.d.). Implementation occurs in three phases: Phase 1 (Years 1–2) focuses on high-traffic corridors linking tram stations to key buildings; Phase 2 (Years 2–4) expands to housing, parking, and Oxygen Park; Phase 3 (Years 4–6) achieves 60–70% campus-wide shade coverage, measured through pedestrian counts and comfort surveys. Scaling can be accelerated through partnerships with national tree-planting initiatives managed by Ashghal using treated wastewater irrigation (Hamad Bin Khalifa University, 2025; Ataullah, 2023).

Alignment with QNV 2030

Park and Plant directly supports NDS3's targets: 25% greenhouse gas reduction, sustainable transportation, and increased urban green space (National Planning Council, 2024). Trees sequester carbon, reduce the urban heat island effect, improve air quality, and create walkable environments that decrease car dependency. Qatar's National Master Plan recommends 60–70% pedestrian shade coverage (Ministry of Municipality and Environment, n.d.), a goal that tree planting directly achieves.

Addressing Limitations

The primary limitation is the time for trees to reach a mature canopy. During the growth period, existing EC canopy structures should be maintained for interim shade. Maintenance costs (irrigation and pruning) are manageable since native species require far less water than ornamental alternatives, and the Qatar Foundation's existing grounds team can absorb these responsibilities.

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